Queue Position Value: Passive Realized Spread at 1-Tick vs Wider Spreads

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ABSTRACT

Using 12.6M touch fills (trades executed at the best quote) in WOOUSDT, RNDRUSDT, MTLUSDT, ASTRUSDT and KEYUSDT (Binance USD-M futures, 2023-09-01 to 2023-10-30), I compare the passive side's profit per fill (passive realized spread, 10-second horizon) across spreads of one to six ticks, using mean profit by spread state with pair-by-day effects, volatility, depth and size-to-depth controls, and day-clustered inference. Passive realized spread at one tick lies 0.10 ticks below the line through its two- and three-tick values, and 0.00 ticks below once volatility, book depth and fill size relative to depth are controlled. A one-tick quote therefore earns no queue-position premium on top of the half-spread it collects, and a maker should not quote at one tick expecting one.

1. Hypotheses

At two or more ticks, a trader can improve the price by one tick and jump ahead of you, so queue position is worth less. At exactly one tick nobody can undercut you by a tick, so queue position should matter more, and passive realized spread at one tick should depart from the smooth line traced by wider spreads.

Q1. Is there a one-tick discontinuity in passive realized spread relative to the benchmark relationship estimated from wider-spread states?

> **H_{01}:** Passive realized spread at one tick lies on the benchmark relationship estimated from wider-spread states (no discontinuity).
> **H_{11}:** Passive realized spread at one tick departs from that benchmark relationship (a discontinuity).

The one-tick state is not random. If one-tick spreads occur in calm markets with deep books and wider spreads in volatile markets with thin books, a difference at one tick could come from those conditions rather than from queue position.

Q2. Is the discontinuity a queue-position effect, or does it reflect volatility and book depth?

> **H_{02}:** After controlling for volatility and book depth, there is no discontinuity at one tick.
> **H_{12}:** After controlling for volatility and book depth, there is a discontinuity at one tick.

2. Data

2.1 Data gathering

Market	Type	Content	Source	Coverage
USD-M futures	bookTicker	Best bid and ask prices and quantities, with timestamps (ms)	Binance data archive	WOOUSDT, RNDRUSDT, MTLUSDT, ASTRUSDT, KEYUSDT; 300 daily files (60 per pair, none missing), 2023-09-01 to 2023-10-30; 137.79M rows
USD-M futures	aggTrades	Aggregated trades: price, quantity, timestamp (ms), buyer-is-maker flag	Binance data archive	The same 300 pair-days; 20.89M rows

Tick size is the smallest price step observed in each pair's quote file: KEYUSDT 0.000001, WOOUSDT and ASTRUSDT 0.00001, RNDRUSDT and MTLUSDT 0.0001.

2.2 Descriptive review

Five pairs, WOOUSDT, RNDRUSDT, MTLUSDT, ASTRUSDT and KEYUSDT, are chosen on these grounds:

- **Band of 40% to 70%** of time at exactly one tick on 2023-09-12 (WOOUSDT 44%, RNDRUSDT 45%, MTLUSDT 61%, ASTRUSDT 67%, KEYUSDT 67%), so that the one-tick state and the two- and three-tick states each carry enough fills to be compared. On six days sampled across the window the share ranges from 33% (WOOUSDT) to 99% (KEYUSDT).
- **Time-weighted share**, to measure how long a maker is in the one-tick state.
- **Window** of 60 consecutive days, 2023-09-01 to 2023-10-30, which gives the 60 day-clusters used for inference.

3. Literature review

This project relies on two studies. Moallemi and Yuan (2017) show that at a one-tick spread, queue position is worth the spread earned minus adverse selection, plus the option value of moving up the queue. They also show that this value falls as the queue lengthens and as price jumps, a component of volatility, become more frequent relative to trades. Yao and Ye (2018) show that a binding tick creates rents for liquidity providers, the side that waits in the queue and whose profit this project measures as passive realized spread, rationed by time priority.

Both studies place the value of queue position at one tick, which is the premise of Q1: passive realized spread at one tick should depart from the line traced by wider spreads. The dependence of that value on queue length and price jumps is the premise of Q2: book depth and volatility can move passive realized spread at one tick, so a gap there can reflect those conditions and not queue position.

4. Methodology

4.1 Model

Equation (1) measures the passive realized spread R for one fill:

$$R_i = s_i \, (p_i - m_{\Delta,i}) \, / \, \tau_p \tag{1}$$

Equation (2) gives the average R at each spread, with the controls x held fixed:

$$R_i = \alpha_{p,d} + \Sigma_{j=2}{}^{6} \, \theta_{p,j} \cdot 1[k_i = j] + \gamma' x_i + u_i \tag{2}$$

4.2 Variables

Symbol	Definition
i, p, d	fill; pair; calendar day
R_i	passive realized spread of fill i, in ticks
s_i	trade direction: +1 buyer-initiated, −1 seller-initiated
p_i	trade price
$m_{\Delta,i}$	mid-quote at the trade time + Δ, Δ = 10 s
τ_p	tick size of pair p
k_i	quoted spread in ticks, (ask − bid) / τ_p, at the last quote before the trade
1[·]	indicator function
$\alpha_{p,d}$	pair-by-day effect
$\theta_{p,j}$	mean of R at k = j minus mean of R at k = 1, pair p ($\theta_{p,1} = 0$)
G	test statistic: R at one tick minus the line through R at two and three ticks, averaged over pairs (4.3, step 3)
x_i, γ	control vector and its coefficients
u_i	error

4.3 Test procedure

1. **Sample.** Fills at the best quote with $k_i \in \{1, \ldots, 6\}$, k_i read from the last quote strictly before the trade. Excluded: deeper levels of a multi-level order, trades with no earlier quote, locked or crossed quotes.

2. **Outcome.** R_i from (1), Δ = 10 s.
3. **Estimation.** R_i from (1) is regressed on (2): Q1 without controls ($\gamma = 0$), Q2 with controls. G = (1/5) Σ_p ($\theta_{p,3} - 2\theta_{p,2}$) is R at one tick minus the line through R at two and three ticks; a line, because the half-spread collected falls 0.5 tick per tick of spread. H_{o1} and H_{o2}: G = 0, with standard errors clustered by calendar day (60 clusters) and rejection at $p < 0.05$. G > 0 implies a queue-position premium.
4. **Controls.** The control vector x_i contains trailing 1-minute volatility and depth at the best quote (deciles within pair), and fill size relative to displayed depth (seven bins).

5. Results

Test	Fills	G (ticks)	p-value	Decision at 5%
Q1: no controls	12.58M	−0.096	0.002	H_{o1} rejected
Q2: with controls	12.56M	−0.004	0.87	H_{o2} not rejected

H_{o1} is rejected at the 1% level (p = 0.002): without controls, passive realized spread at one tick lies 0.096 ticks below the line through its two- and three-tick values.

H_{o2} is not rejected at the 5% level (p = 0.87): with volatility, depth and fill size controlled, the gap is −0.004 ticks.

Robustness. Both tests are re-estimated at a 1-second horizon and, at the 10-second horizon, pair by pair. Q2 keeps its decision in both. Q1 keeps its decision at 1 s but not pair by pair, because only ASTRUSDT rejects, so Q1 is the weaker result.

6. Interpretation

Passive realized spread at one tick shows no queue-position premium. H_{o1} is rejected, but profit per fill at one tick is lower than wider spreads imply, not higher. H_{o2} is not rejected: once volatility, depth and fill size are controlled, that difference disappears, so it reflects market conditions and not queue position. Market makers and other liquidity providers who choose between joining the one-tick queue and quoting wider therefore earn no extra profit per fill from queue position at one tick.

Limits:

- Five pairs over 60 days in 2023, touch fills only: the results apply to trades at the best quote in these markets.
- Realized spread is before fees and rebates, which are part of what a passive quote earns.
- The one-tick benchmark is a straight line through the two- and three-tick values; a different benchmark shape would change G.

7. Sources

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